

Nguyen Thanh Long

Backend / Platform Engineer

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SUMMARY

Backend engineer, 2.5 years, currently the only engineer on the cloud side of an IoT startup: the NestJS and MongoDB APIs, the MQTT-to-Kafka pipeline the devices talk through, and the Kubernetes and AWS it runs on. Most of what I know came from running that system, not only from adding features to it.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, C#, Python, SQL
Backend	Node.js, NestJS, Express.js, ASP.NET Core, REST, WebSockets
Data & messaging	MongoDB (Atlas and self-hosted replica sets), PostgreSQL, Redis, MySQL, DynamoDB, Apache Kafka (Strimzi), MQTT (EMQX)
Infrastructure	Kubernetes (k0s, kubeadm, KEDA), AWS (ECS Fargate, S3, DynamoDB, Lambda, CloudWatch), Docker, Terraform, ArgoCD, Nginx
Operations	Prometheus, Grafana, Loki, GitHub Actions, GitLab CI, Jest
Auth	Firebase Auth, Keycloak, JWT, ABAC, OAuth 2.0

EXPERIENCE

Backend / Platform Engineer — Rogo Solutions JSC

Jan 2024 – Present · Hanoi

A ~20-person IoT startup. I am the only engineer on the cloud side, so the API, the message pipeline, the clusters and the deploy path are all mine. It serves 640k users and 69k devices for 26 partner projects, at roughly 2.6M API requests a day and 15.5k concurrent MQTT connections, on 3 Kubernetes clusters and AWS ECS.

- I run the pipeline that carries device state. Partner traffic used to go through one Kafka broker with no replication, so I moved it onto a replicated three-broker cluster with a separate listener for partner systems. The consumer merges repeated state updates instead of writing every one, and scales on consumer lag rather than a fixed replica count.
- One deployment serves 26 partner projects, so most feature work turns out to be access control. I own the authorization layer that decides which partner can see which devices and locations, and moved things that used to be global, like the SMS provider, to per-project configuration.
- There was no monitoring when I started. Prometheus, Loki and Grafana now cover the three clusters with the AWS side pulled in from CloudWatch, so an incident starts at a dashboard instead of a guess. I also own the staging-to-production path across four repositories.
- Most of my performance work has been finding the cause rather than writing anything clever. A device query took eleven minutes because nothing was indexed behind a \$lookup, and the plan had been to pay for a bigger database tier instead. Auth middleware was verifying a token signature on every request, which no APM was showing. A partner's SDK sent everything at once on the hour. Each fix was small once the cause was clear.
- The test suite had been abandoned because running it exhausted memory and froze people's machines, so nothing was gated on it. Moving to Jest 29 with transpile-only tests and a separate type-check brought it back to a few seconds, and back into CI.
- I worked out whether the product could be sold as an on-premise install: what replaces the managed pieces (Keycloak, MongoDB and MinIO for Firebase, DynamoDB and S3), how existing accounts migrate without forcing password resets, and enough of it built to boot. It has not shipped. What the exercise produced was a concrete list of what still blocks it.

SELECTED PROJECTS

- IoT Cloud MCP** — TypeScript, Python, MCP SDK
An MCP server for IoT cloud platforms, so an assistant can query and control devices through one interface rather than a separate integration per platform.
- Novel Studio** — NestJS, PostgreSQL + pgvector, LLM
A long-form writing tool. The awkward part is that prompt cost grows with the story, so it keeps a canon store, a rolling summary and vector retrieval instead of resending history.
- MathSketch Tutor** — NestJS, mathjs, Firebase, LLM
Feedback on a handwritten maths problem. A computer-algebra engine decides whether a step is right and the model only writes the explanation, so it cannot invent a correction that is wrong.

EDUCATION & ADDITIONAL

- B.Sc. in Software Engineering — FPT University**

2022 – 2026

Final-year project: a real-time multiplayer game backend (ASP.NET Core, WebSockets, MySQL) using lag compensation to keep play fair across uneven network conditions.
- Certifications:** Project Management Principles (UC Irvine, 2025) · Software Development Lifecycle Specialization (University of Minnesota, 2024).
- Coach competitive-programming students for national Informatics examinations, weekly since 2024; freelance backend work alongside full-time employment. English: professional working proficiency.